F O R M 2

THE PATENTS ACT, 1970

(39 of 1970)

And

THE PATENTS RULES 2003

COMPLETE SPECIFICATION

(See Section 10; Rule 13)

Smart Notification Wristband for Sensory-Impaired Individuals

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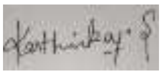
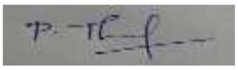
THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES THE INVENTION AND THE MANNER IN WHICH IT IS TO BE PERFORMED:

APPLICANT NAME AND SIGNATURE

Signature:  



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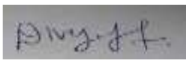
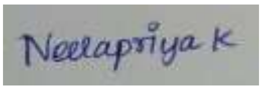
Signature:   

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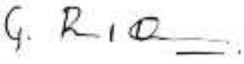
Signature: 

Name: 7. Dr. G .Tamilselvan.

Title of the invention:

Smart Notification Wristband for Sensory-Impaired Individuals

Field of the invention:

The present invention relates to assistive wearable technology for sensory-impaired individuals. It specifically addresses real-time alert systems using visual and haptic feedback. The invention enhances safety by notifying users of environmental hazards in household settings.

Prior art to the invention:

Patent document with application number US20150230461A1, titled – “Wearable Notification System for the Hearing Impaired”, disclosure – “The invention describes a wrist-worn device that receives wireless alerts from doorbells, fire alarms, or other household devices and converts them into vibratory signals. The system focuses on sound-based event detection and transmission, alerting the user through vibration only. However, it lacks an integrated OLED display for visual alerts, does not incorporate multiple environmental sensors such as gas leak or electrical fault detectors, and offers no health monitoring features. Furthermore, it does not provide advanced feedback mechanisms such as repeated alerts on user inactivity or multi-type message differentiation.” COPYRIGHTUSPTO2015 Reference numerals] (100) Wearable device; (110) Vibration module; (120) Wireless receiver; (130) Sound-based event detector.

Objects of the invention:

The primary object of the invention is to provide a smart wearable device that alerts sensory-impaired individuals to household hazards.

Summary of the invention:

The invention is a smart wristband that provides real-time alerts to sensory-impaired users via vibration and OLED display. It receives signals from a home-based transmitter connected to safety sensors. The device also monitors basic health parameters for enhanced user safety.

Detailed description:

The Smart Notification Wristband for Sensory-Impaired Individuals is a wearable device designed to enhance the safety and independence of people who are deaf or have impaired sensory perception, particularly in home environments. This invention integrates advanced alerting and health-monitoring features into a compact wrist-worn form, enabling users to stay aware of important household events that would typically rely on sound or smell. The wristband functions as a receiver in a two-unit system, where a separate home-based transmitter detects specific conditions through a series of sensors and communicates alerts wirelessly to the wearable device.

The transmitter is powered by a 12V, 3500mAh rechargeable battery, ensuring continuous operation even during power outages. It is equipped with multiple sensors including a doorbell interface that senses visitor presence, a float switch to detect water tank overflow, a gas sensor to identify the presence of harmful gas leaks, and an earth leakage sensor that detects electrical faults. These sensors are critical for preventing common domestic hazards, but traditional alert systems often rely on auditory alarms or olfactory signals, which are ineffective for individuals with hearing or smell impairments. The transmitter converts these sensor signals into RF (radio frequency) signals and transmits them to the receiver unit.

The wristband, acting as the receiver, is designed for comfort and constant wearability. It incorporates a 0.96-inch OLED display that provides clear visual alerts, a Pro Mini microcontroller based on the Atmega328 platform for processing the incoming data, and a vibration motor that delivers haptic feedback to the user. It is powered by a 3.7V, 1250mAh lithium battery that supports extended usage. When an alert signal is received, the wristband decodes it and presents a corresponding message on the display—such as “DOOR OPEN” for a doorbell press, “WATER TANK” for water overflow, “ECL” for an electrical fault, or “GAS” for a gas leak. Simultaneously, the vibration motor is activated for three seconds to physically notify the wearer. If no user interaction is detected in response to this initial alert, the vibration is repeated at a higher frequency to ensure the message is acknowledged.

Beyond its primary safety functions, the wristband also incorporates health-monitoring features, including a heart rate sensor, a body temperature sensor, and a blood oxygen sensor. These additions allow the device to provide real-time biometric feedback, further supporting the well-being of the wearer, especially for elderly users who may require regular health monitoring. The combination of safety and health functionalities in a single device makes this wristband a comprehensive solution for personal care in a home setting.

The Smart Notification Wristband is particularly advantageous for individuals who are deaf or sensory-impaired, offering them a reliable, non-intrusive method to stay informed about critical home events. Its reliance on visual and tactile signals rather than auditory or olfactory cues ensures inclusivity. While it is tailored for the disabled community, it also serves as an advanced safety and health tool for general users, demonstrating its broad applicability and value as a modern assistive technology.

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Claims:

1. A wearable notification device comprising an OLED screen, microcontroller, RF receiver, and a vibration motor, designed to receive and alert home-based signals.

2. The system wherein the transmitter comprises sensors for detecting doorbell rings, water overflow, gas leakage, and electrical leakage.

3. The receiver unit configured to respond to specific inputs with unique display messages and vibration patterns.

4. A system wherein lack of user interaction triggers repeated alerts.

5. The device as claimed above, further integrating BPM, oxygen, and temperature sensors to monitor user health in parallel.

Abstract:

The present invention relates to a smart wearable notification device, specifically designed for individuals with hearing and olfactory impairments. The system comprises a wearable wristband integrated with an OLED display, sensors, RF communication, and haptic feedback, capable of receiving alerts from various household safety and utility systems. The device ensures safety and awareness for individuals who cannot perceive auditory or smell-based cues by delivering clear visual and tactile alerts. It is particularly helpful for deaf individuals, elderly persons, and general users.